Hints: Limits – 6. Asymptotics and sharp estimates: sums, integrals, Stirling

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For monotone f compare $f(k)$ with $\int_k^{k+1} f$ on each unit interval: for increasing f one has $f(k) \leqslant \int_k^{k+1} f \leqslant f(k+1)$. Summing gives $\sum_{k=1}^{n-1} f(k) \leqslant \int_1^n f \leqslant \sum_{k=2}^{n} f(k)$, so the difference $\sum_{k=1}^{n} f(k) - \int_1^n f$ lies between $f(1)$ and $f(n)$.

Hint 2. For decreasing $f \geqslant 0$ put $d_n = \sum_{k\leqslant n} f(k) - \int_1^n f$. Then $d_n - d_{n+1} = \int_n^{n+1} f - f(n+1) \geqslant 0$ and $d_n \geqslant f(n) \geqslant 0$, so d_n converges. With $f(x) = 1/x$ this gives $\gamma = \lim(H_n - \ln n)$.

Hint 3. For the next term write $H_n - \ln n - \gamma = \sum_{k\geqslant n} \bigl(\ln(1 + \frac{1}{k}) - \frac{1}{k+1}\bigr)$ and expand: the summand is $\frac{1}{2k^2} + O(k^{-3})$. Since $\sum_{k\geqslant n} k^{-2} = \frac{1}{n} + O(n^{-2})$, you get $H_n = \ln n + \gamma + \frac{1}{2n} + O(n^{-2})$.

Hint 4. Numerically: $H_{10} = 2.928968\ldots$, $\ln 10 + \gamma = 2.879801\ldots$. The two-term approximation $\ln n + \gamma$ is off by about $0.049 \approx \frac{1}{2n}$; adding $\frac{1}{20}$ leaves an error of about $-0.00083 \approx -\frac{1}{12n^2}$.

Problem 2

Hint 1. Take $f = \ln x$ and compare $\ln n! = \sum \ln k$ with the trapezoid rule rather than with the bare integral: consider $d_n = \ln n! - (n + \frac{1}{2}) \ln n + n$.

Hint 2. Compute $d_n - d_{n+1} = (n + \frac{1}{2}) \ln \frac{n+1}{n} - 1$. With $t = \frac{1}{2n+1}$ one has $\frac{n+1}{n} = \frac{1+t}{1-t}$, so $d_n - d_{n+1} = \frac{1}{2t} \ln \frac{1+t}{1-t} - 1 = \frac{t^2}{3} + \frac{t^4}{5} + \ldots$

Hint 3. From this, $0 < d_n - d_{n+1} < \frac{1}{3} \sum_{j\geqslant 1} t^{2j} = \frac{1}{12}(\frac{1}{n} - \frac{1}{n+1})$. So d_n decreases and $d_n - \frac{1}{12n}$ increases; both converge to some C. This gives $n! = e^C \sqrt{n}\,(n/e)^n e^{\theta_n}$ with $0 < \theta_n < \frac{1}{12n}$. (The lower bound $\frac{1}{12n+1}$ needs a slightly sharper estimate of the series from below.)

Hint 4. For the constant use Wallis: with $a_m = \int_0^{\pi/2} \sin^m x\, dx$ one has $a_{2n}/a_{2n+1} \to 1$, which gives $\frac{\pi}{2} = \lim \frac{2^{4n}(n!)^4}{((2n)!)^2(2n+1)}$. Substitute $n! \sim e^C \sqrt{n}(n/e)^n$ and solve to get $e^C = \sqrt{2\pi}$.

Problem 3

Hint 1. Write $S_n = \frac{1}{n} \sum_{k=1}^{n} f(k/n)$ with $f(x) = \frac{1}{\sqrt{1+x^2}}$. This is a Riemann sum, and $\int_0^1 f = \operatorname{arsinh} 1 = \ln(1 + \sqrt{2})$.

Hint 2. For the second limit you need the error of the right-endpoint Riemann sum of a smooth f. On $[\frac{k-1}{n}, \frac{k}{n}]$, expanding f around the right endpoint gives $\frac{1}{n} f(\frac{k}{n}) - \int f = \frac{1}{2n^2} f'(\frac{k}{n}) + O(n^{-3})$. Summing gives $\frac{f(1)-f(0)}{2n} + O(n^{-2})$.

Hint 3. So $n(S_n - \ln(1 + \sqrt{2})) \to \frac{f(1)-f(0)}{2} = \frac{1}{2}(\frac{1}{\sqrt{2}} - 1) = \frac{\sqrt{2}-2}{4}$.

Problem 4

Hint 1. Let $I_m = \int_0^1 \frac{dx}{1+x^m}$. Show that $\frac{1}{2} < I_m < 1$. Then $c_m = (1/I_m - 1)^{1/m}$ is forced, and it lies in $(0, 1)$.

Hint 2. You need $k c_k^k = k\, \frac{1-I_k}{I_k}$. Write $1 - I_k = \int_0^1 \frac{x^k}{1+x^k} dx$ and substitute $t = x^k$.

Hint 3. You get $1 - I_k = \frac{1}{k}\int_0^1 \frac{t^{1/k}}{1+t}\,dt$, and the integral tends to $\int_0^1 \frac{dt}{1+t} = \ln 2$, while $I_k \to 1$. The limit is $\ln 2$.

Problem 5

Hint 1. Use the first step of Euler–Maclaurin from Theorem 2: $\sum_{k=1}^{n} f(k) = \int_1^n f + \frac{f(1)+f(n)}{2} + R_n$ with $R_n = \int_1^n \frac{\{x\}(1-\{x\})}{2} f''(x)\,dx$. If $\int_1^\infty |f''| < \infty$, then $R_n = R_\infty + o(1)$, and everything constant goes into one constant C.

Hint 2. For $f = \sqrt{x}$: $f'' \sim x^{-3/2}$ is integrable, so $\sum_{k\leqslant n} \sqrt{k} = \frac{2}{3}n^{3/2} + \frac{1}{2}\sqrt{n} + C + o(1)$. The terms used are the integral, the boundary term $f(n)/2$ and the convergent remainder. The next term, $\frac{B_2}{2} f'(n) = \frac{1}{24\sqrt{n}}$, is already $o(1)$. (In fact $C = \zeta(-\frac{1}{2}) \approx -0.2079$.)

Hint 3. For $f = \ln^2 x$: $\int_1^n \ln^2 x\,dx = n\ln^2 n - 2n\ln n + 2n - 2$, and $f'' = \frac{2-2\ln x}{x^2}$ is integrable. So $\sum_{k\leqslant n} \ln^2 k = n\ln^2 n - 2n\ln n + 2n + \frac{1}{2}\ln^2 n + C' + o(1)$. (In fact $C' = \zeta''(0) \approx -2.0064$.)